

Supporting Information

Instructions for Growing a Display Crystal of Recycled Potassium Tris(oxalato)ferrate(III) Trihydrate

Peter C. Junk, College of Science and Engineering, James Cook University, Townsville, Qld, 4811 Australia

John Bacsá, Chemistry Department, Emory University, Atlanta, GA 30322 United States

Nicholas C. Thomas, Chemistry Department, Auburn University at Montgomery, Montgomery, AL 36124-4023 United States

Recrystallization of student potassium tris(oxalato)ferrate(III) trihydrate waste

100g of student waste material is added to 200 mL deionized water and heated to about 45 °C with stirring to dissolve the solid. The warm solution is filtered and the filtrate allowed to cool for 48-hrs. The pure potassium tris(oxalato)ferrate(III) trihydrate is collected by filtration, air dried, and weighed (65.5 g). The filtrate may be left for several more days and an additional quantity of the green product will deposit. Total yield: 74.4 g

Growing the potassium tris(oxalato)ferrate(III) trihydrate crystal

The above potassium tris(oxalato)ferrate(III) trihydrate is added to 500 mL of deionized water in a 600 mL beaker which is stirred and heated slightly (as above) to dissolve the solid. The solution is filtered and returned to the clean beaker then left uncovered in a secure location. Crystals will initially deposit over several days, and a seed crystal large enough to tie a thread around should be removed.

The end of a 30-cm length of nylon thread is tied around the seed crystal obtained from the saturated solution (cotton thread can be used but may degrade and break over time; nylon thread is more resilient but can be challenging to tie unless the seed crystal is at least 5 mm long). The other end of the thread is tied to and wrapped around a popsicle stick (or pencil or spatula) and the crystal is suspended back in the saturated green solution.

Growth will be rapid initially, then slow down as the crystal surface area increases. Periodically, as additional crystals deposit on the bottom of the beaker, the growing crystal should be removed and the solution filtered to remove deposits (which can be recycled) and the crystal returned to the solution. This can be continued for many weeks until the desired crystal size is achieved. However, the process can be continued indefinitely by preparing more saturated solution and continually refilling the beaker and re-suspending the crystal (and eventually moving to a larger beaker if necessary).

Note that room temperature fluctuations may affect the crystal (a temperature increase could partially dissolve the crystal). Also, if the crystal growth is too rapid, small irregular outgrowths may appear on the otherwise smooth crystal surface. These can be removed by carefully sanding

away with fine sandpaper. But care should be taken when handling the delicate crystal as it can easily break.

When the crystal growth is completed (5 weeks for the crystal shown), the thread is cut off, the crystal dried, and coated with clear nail varnish or lacquer spray (best) for protection. We have preserved crystals in this manner for many years with little surface degradation (eg due to moisture loss from the crystal lattice). The final crystal mass based on the 74.4 g of recrystallized student waste used was 38.3 g or 51%. However, larger crystals can be prepared if more student waste is used. Gloves should be worn when handling or displaying the crystal.

All unused solid potassium tris(oxalato)ferrate(III) trihydrate can be returned to the student waste container and solutions can be left to evaporate to dryness and similarly stored. The recycled material can be recrystallized and reused many times.